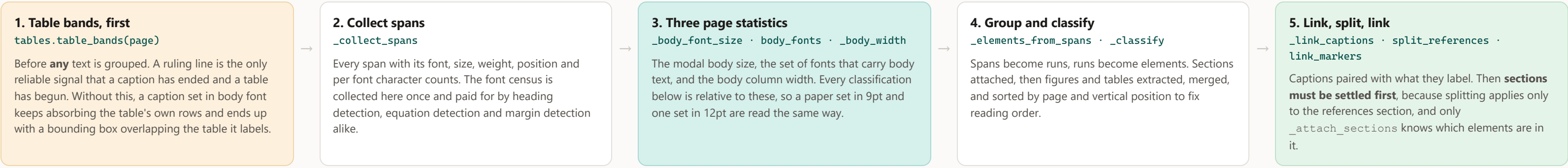


Extraction pipelines, one per kind

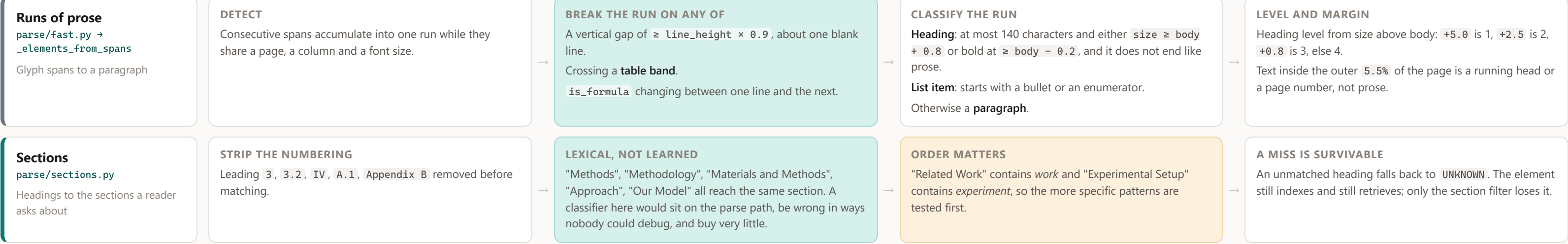
What happens inside the single box marked **Parse** on the architecture sheet. A PDF has no paragraphs, no tables and no figures in it: it has glyphs at coordinates, ruling lines, and image objects. Everything below is the work of turning that back into things a reader named. Every constant shown is the real value from the module beside it.

0 THE ORDER THE FAST PARSER DOES THINGS, AND WHY THAT ORDER

`FastParser.parse()`. The sequence is not arbitrary: three steps exist only because a later step needs their output, and moving them breaks things that look unrelated.

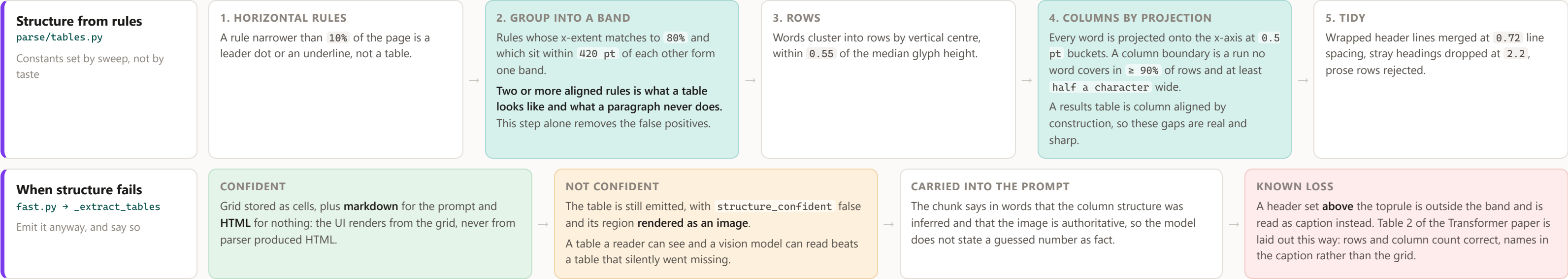


1 TEXT: PARAGRAPHS, HEADINGS AND LIST ITEMS



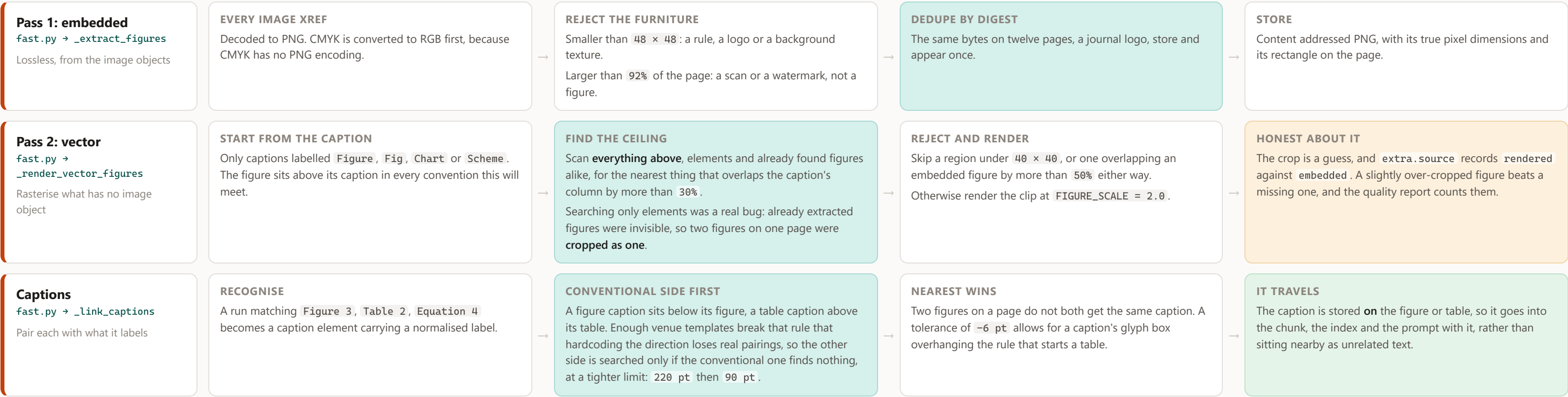
2 TABLES, THE HARDEST ONE

Scientific papers use **booktabs**: rules above the header, below the header and at the foot, and no vertical rules at all. That breaks every finder that looks for intersecting lines, and it breaks the usual fallback of inferring both axes from whitespace. Both failures were measured before this was written: pdfplumber and PyMuPDF's own `find_tables` each turned body text on page 3 of "Attention Is All You Need" into a **22 by 8 table of word fragments**.



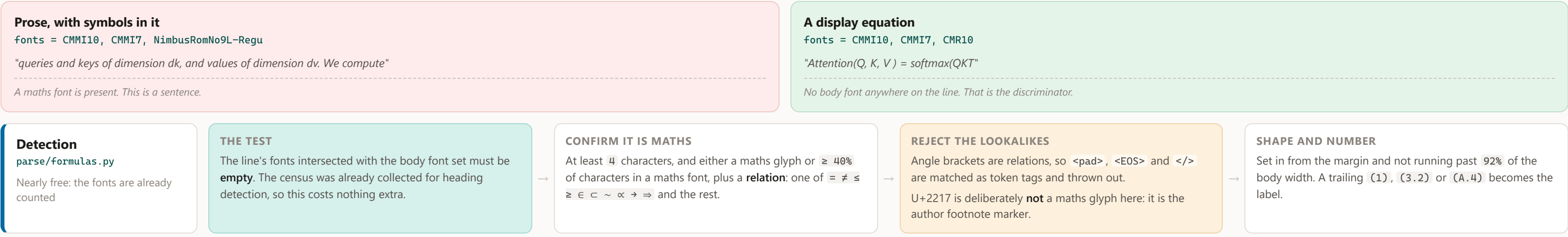
3 FIGURES, TWO PASSES BECAUSE THERE ARE TWO KINDS

A photograph is an embedded image object and can be extracted losslessly. A chart drawn with vector primitives, which is **most charts matplotlib or TikZ produce**, has no image object at all and has to be rasterised from the region a caption points at.



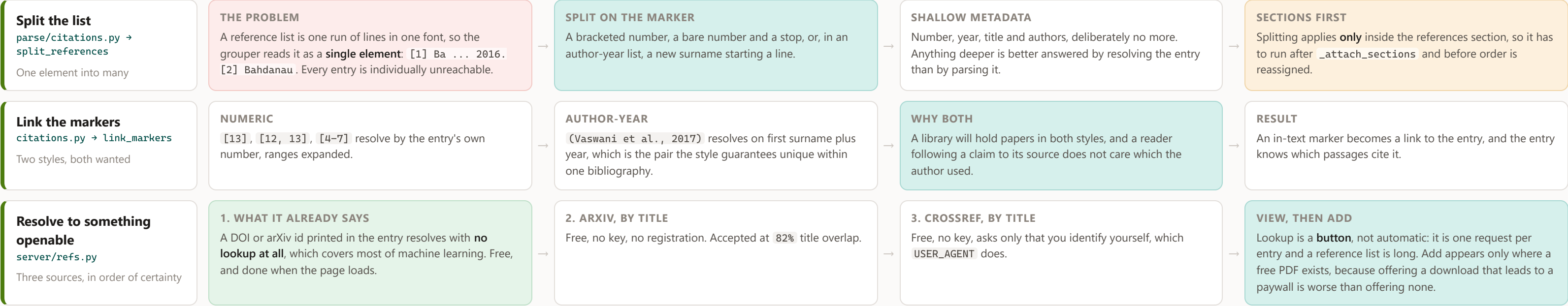
4 DISPLAY EQUATIONS, FOUND BY WHAT IS ABSENT

The obvious test, "does this line use a maths font", does not work: in a LaTeX paper the body font and the maths fonts are mixed constantly inside ordinary prose, because every inline symbol is set in one. Both lines below are from page 4 of "Attention Is All You Need".

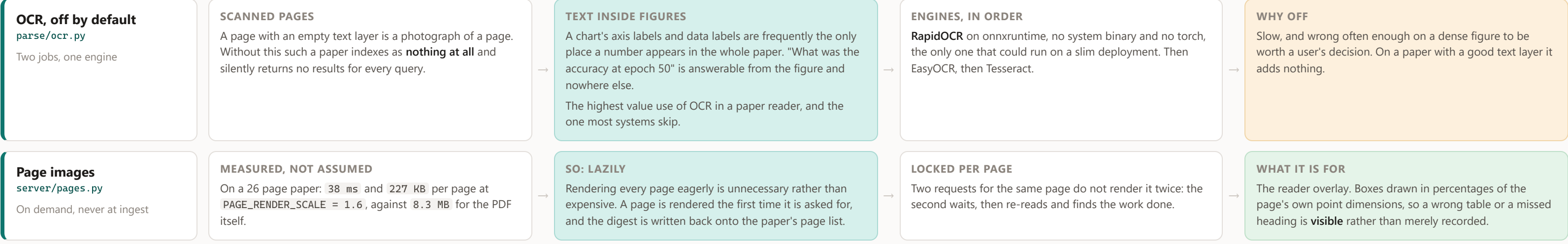


Two equations do not merge into one. Consecutive formula lines are allowed a looser vertical gap, `line_height` $\times 2.5$ rather than $\times 0.9$, because a multi-line equation is set with wide leading and the ordinary gap test splits one equation into three. What this does **not** do is recover LaTeX: getting `\frac{QK^{\text{T}}}{\sqrt{d_k}}` back from glyph positions is a research problem, and the deep tier's formula enrichment is the right tool. The text, the box and the number are enough to find it, cite it, show it and hand the region to a vision model.

5 REFERENCES AND THE CITATIONS THAT POINT AT THEM



6 OPTIONAL PASSES: OCR, AND RENDERING A PAGE



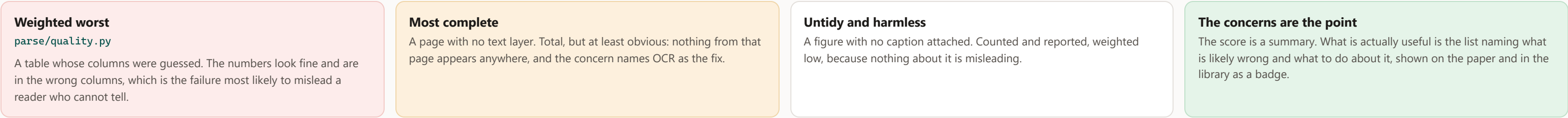
7 WHAT EACH KIND BECOMES DOWNSTREAM

One table holds all seventeen kinds, and the differences show up in how each is chunked and what the lexical index is allowed to see. `Element.search_text()` returns label, caption, text, every table cell, and a figure's OCR text and description, so a number that exists only in a table cell or only on a chart axis is still findable by typing it.

KIND	CHUNKED AS	WHAT THE LEXICAL INDEX SEES	ATTACHED TO A PROMPT AS	WHERE YOU SEE IT
Paragraph, list item	Merged to a 450 token budget, 80 overlap, split on sentences	Its text	Numbered excerpt	Reader, All Views, evidence cards
Heading	Prepended to every chunk beneath it as the section path	Its text	Part of each excerpt's context line	Outline, section filter
Table	Whole, never split, as markdown plus its caption	Every cell, via <code>cell_text()</code>	Markdown, plus a warning when structure was inferred	Rendered from its grid, and as an image when unconfident
Figure	Caption, OCR text and description. A figure with none of those still becomes a chunk	Caption, OCR text, description	The image itself , on providers that accept one	Figure gallery, reader overlay, evidence
Formula	Atomic, with its number as a label	Its text and label	Excerpt, and the region as an image	Equations view, reader overlay
Reference	Atomic, one chunk per entry	Authors, title, year, venue	Excerpt, when the question is about prior work	References tab, with View and Add
Caption	Not chunked alone: it lives on the figure or table it labels	Through its target	Through its target	Under the figure, and as the label everywhere
Page header, footer	Not chunked. Kept, not indexed as prose	Excluded from reading order	Never	Reader overlay only, so a parse can be checked

8 HOW MUCH TO TRUST ONE PARSE

The parser already knows how well it did. It records whether a table's columns were read from ruling or inferred, whether a figure was an image object or a cropped guess, whether a page had a text layer, and whether a caption was ever matched. **None of that was ever shown**, so a bad parse looked exactly like a good one and a wrong number in a table looked as authoritative as a right one.



9 THE DEEP TIER, WHERE IT IS AVAILABLE

Docling was chosen over Marker and MinerU **deliberately, and not because it wins on raw accuracy**. On olmOCR-bench as of 2026 Marker 2 leads at 76.0, MinerU is at 72.7 and Docling's non-VLM pipeline is at 50.3.

